

Class 4

Data Cleaning, Topology and Generalization Techniques

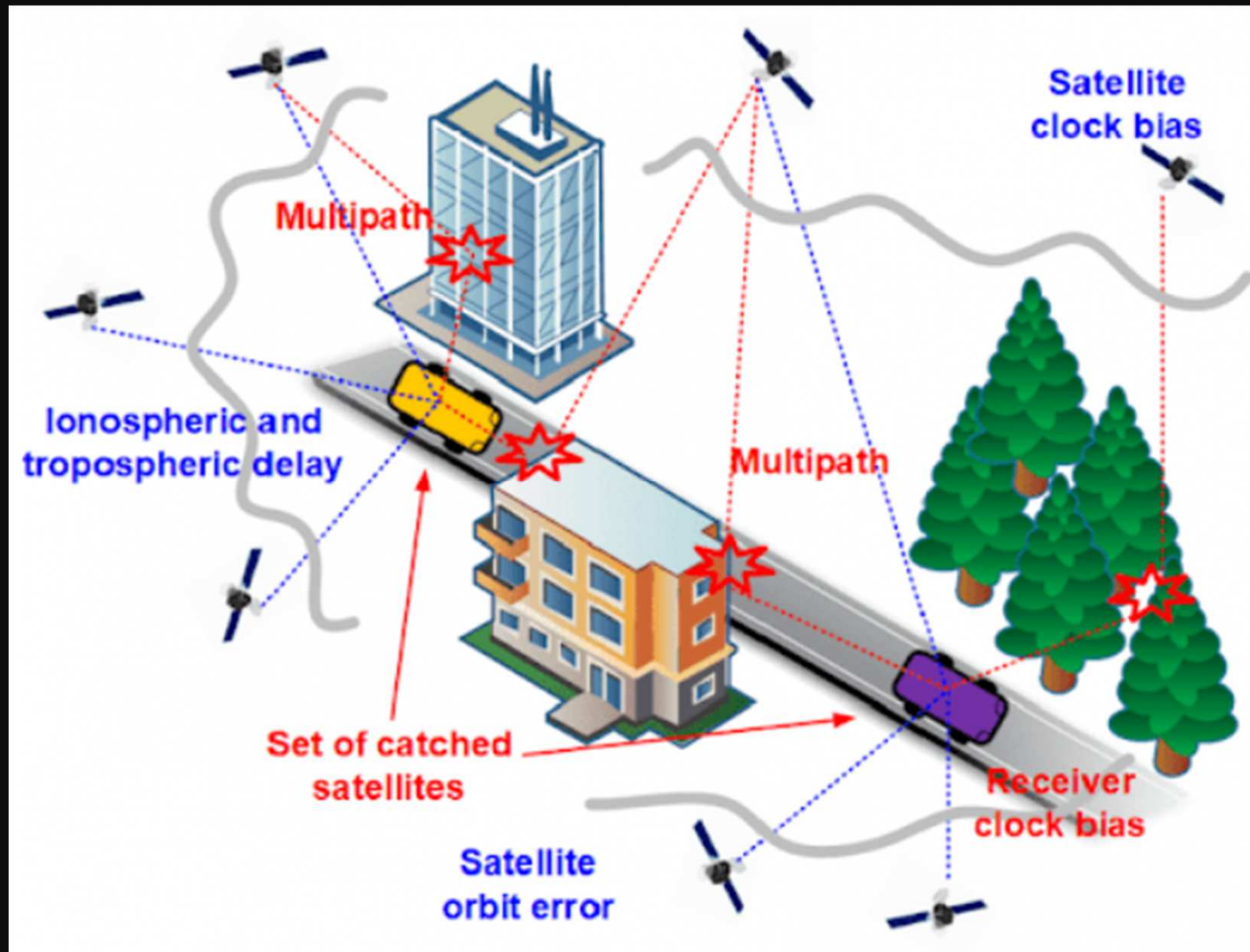
Dr. P. Rama Chandra Prasad
Lab for Spatial Informatics, IIIT, Hyderabad

August 28th, 2026

Data editing

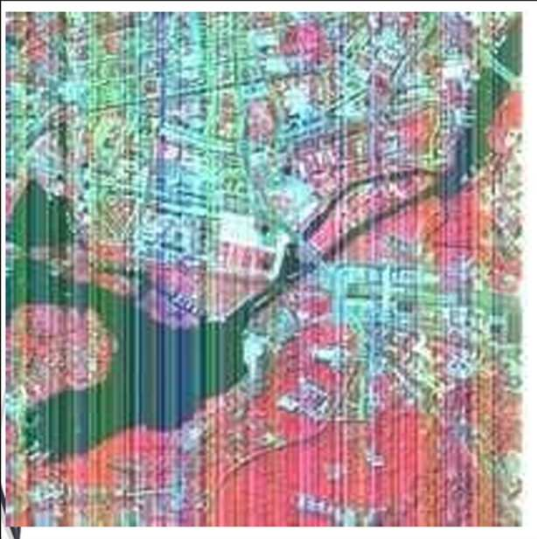
Primary data capture

1. Positional accuracy of GPS receivers



2. Device/instrument operator/malfunction of electronic scanners

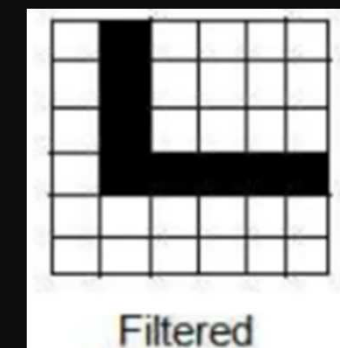
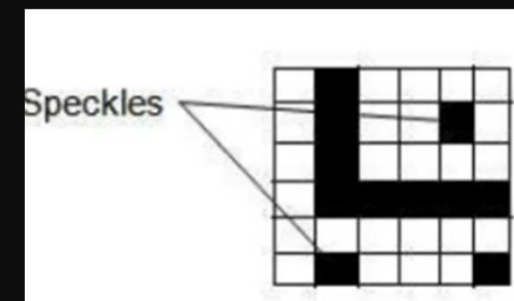
Systematic Noise



Random Noise

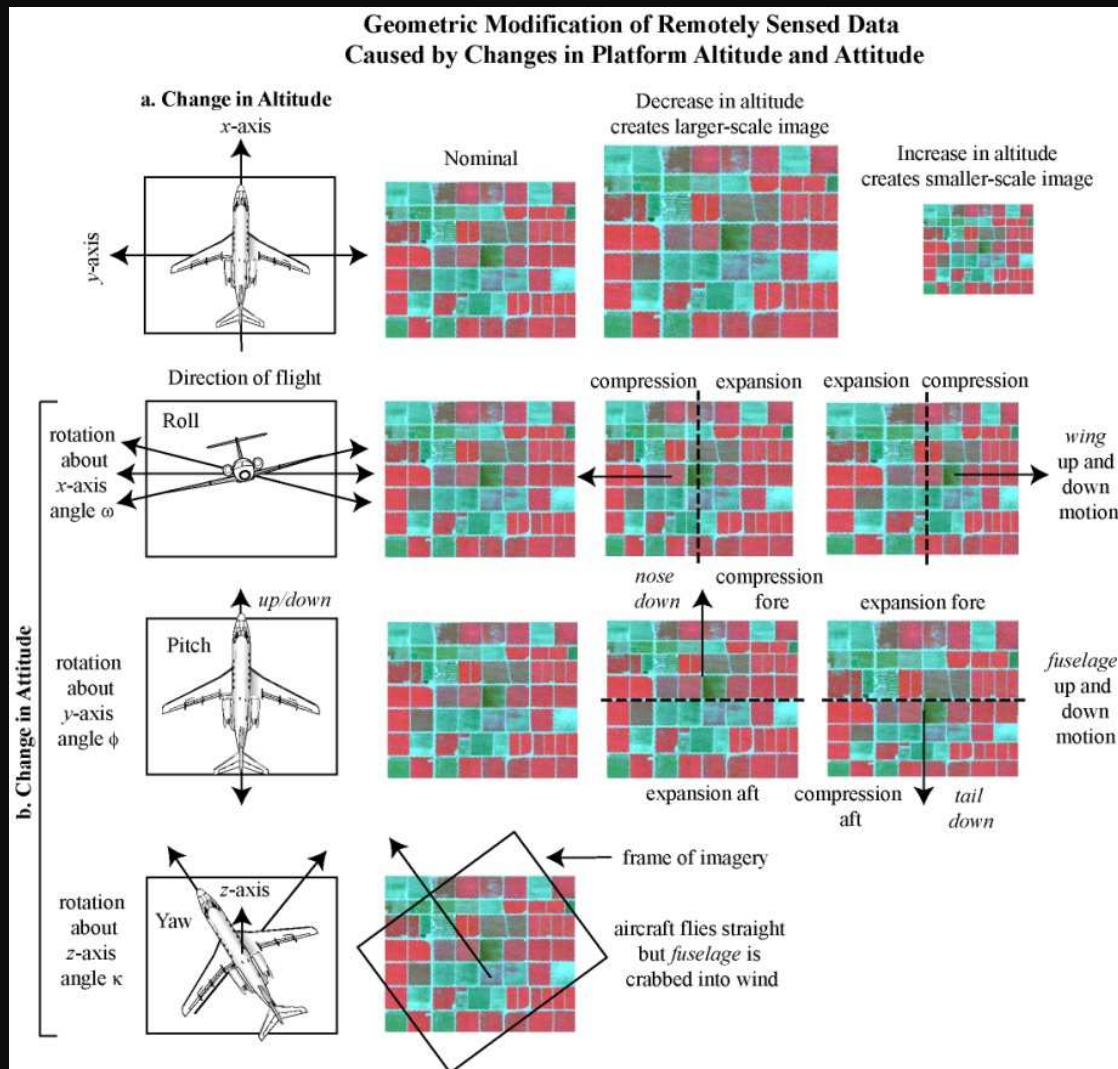


Line dropouts

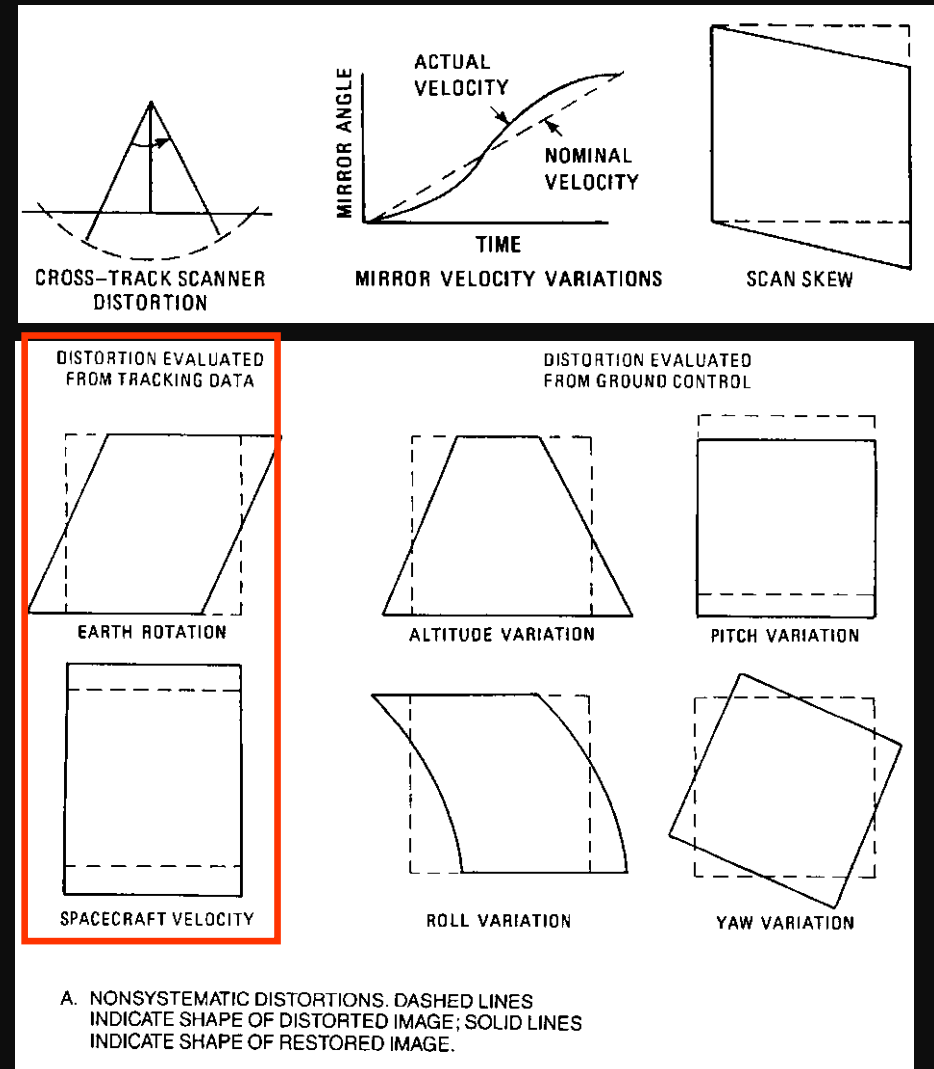


3. Geometric Distortion in Remote Sensing Data

Aerial Photography



Satellite imagery

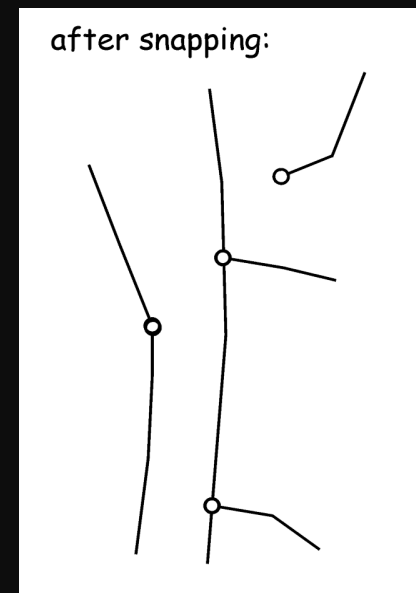
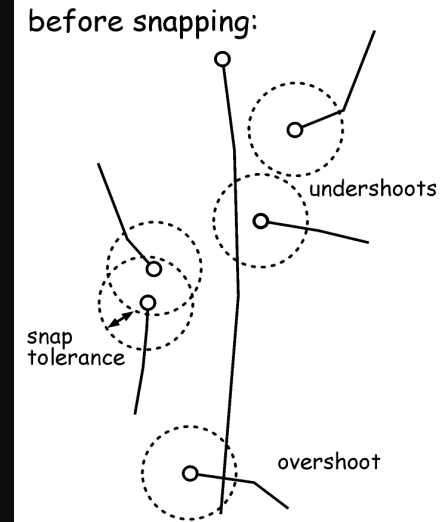
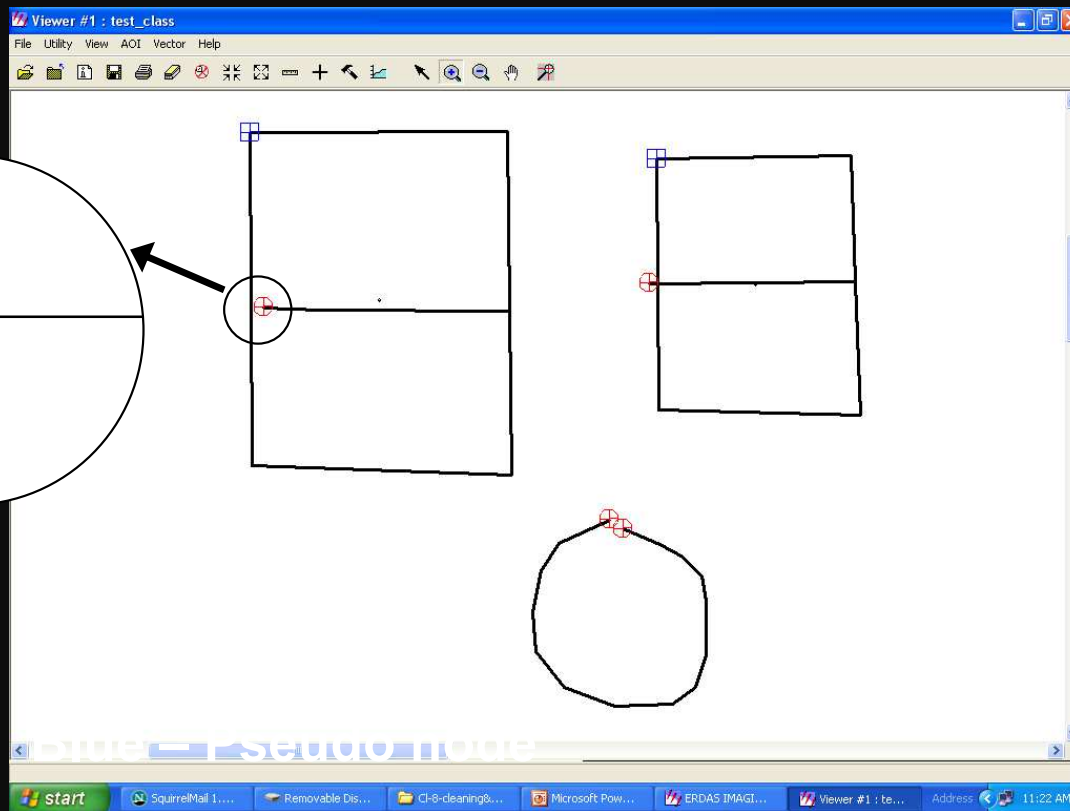


Secondary data capture

1. Digitizing with no proper use of editing tool

Entity errors

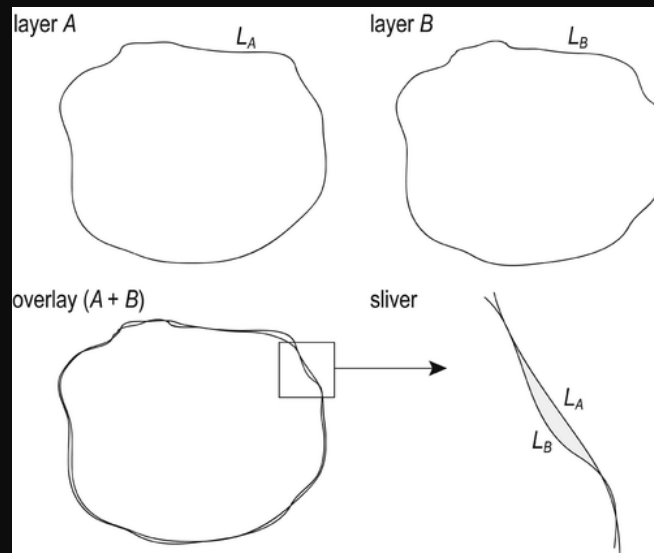
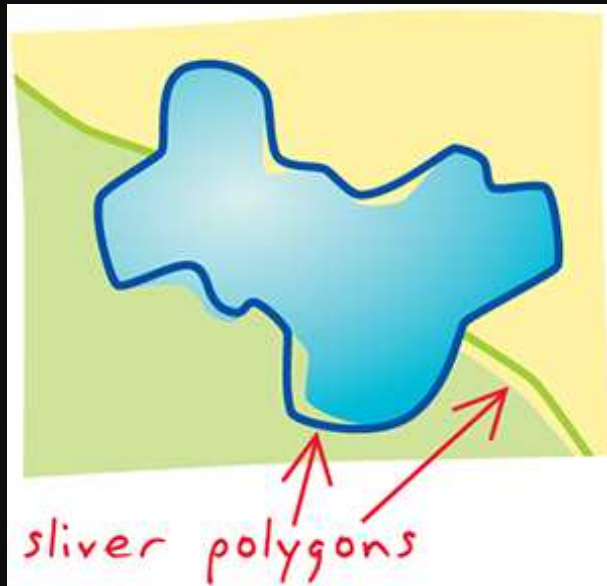
Node errors



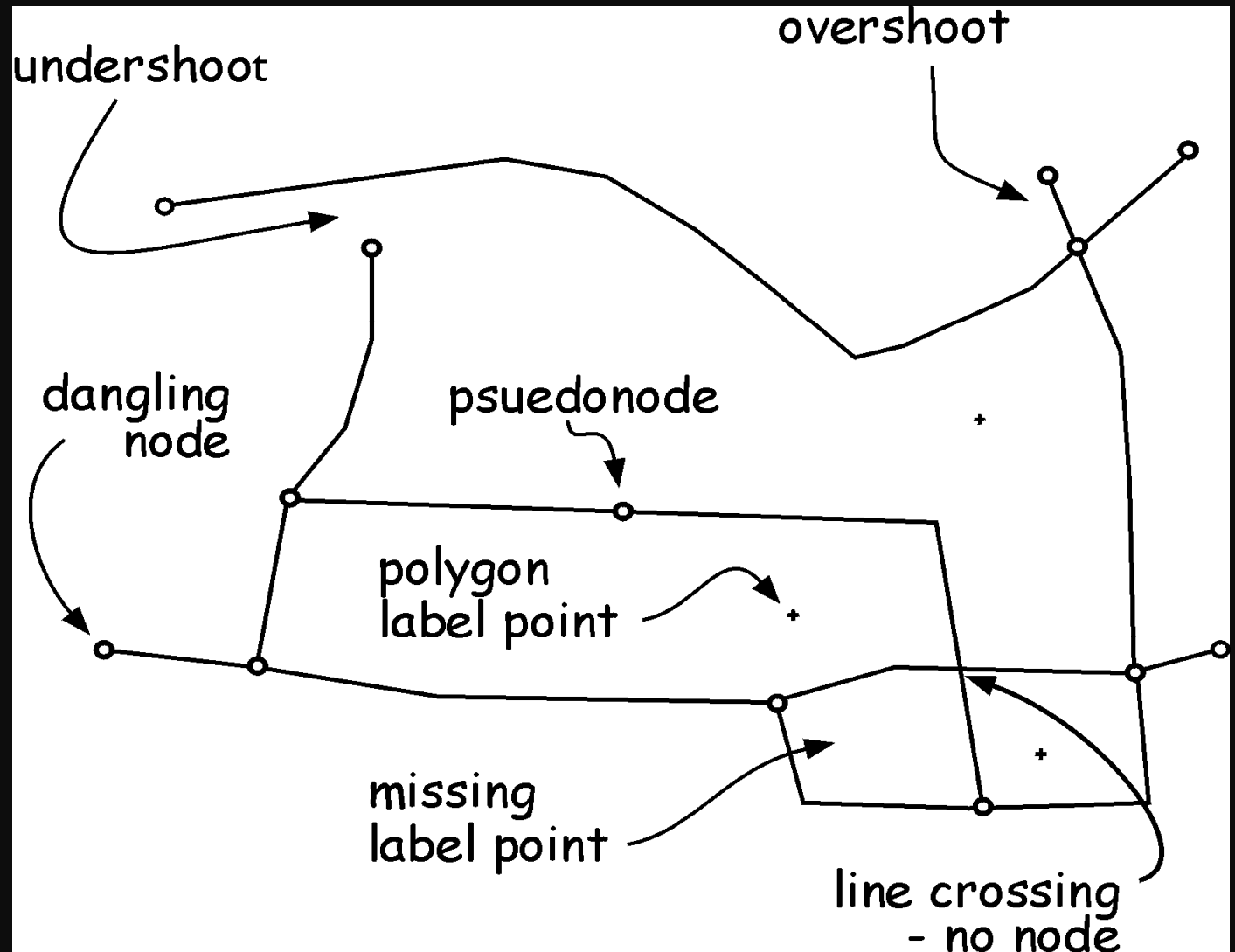
Interactive editing in a proper snapping environment
Automated data **cleaning (topology)**

Why to fix node errors
It validates connectivity
It validates contiguity

Polygon errors



Editing the data

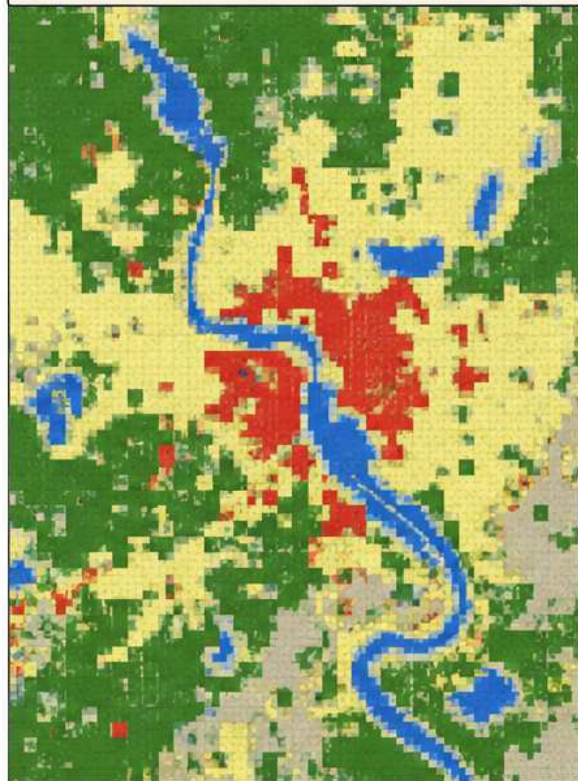


Add missing arc; Correct overshoot / under shoot
Fix an open polygon, Split & unsplit, Move & add vertex, Labels – Add, delete or change

2. Vectorization without post processing

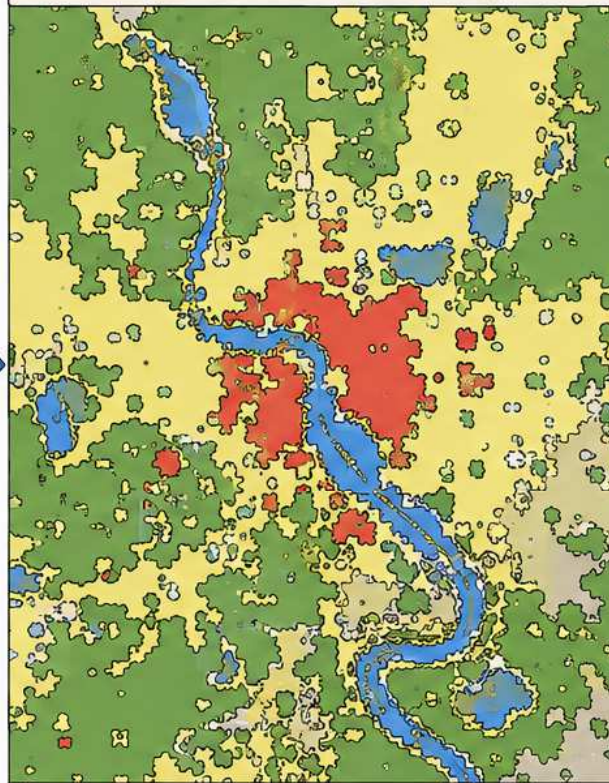
Vectorization Without Post Processing

1. Source Raster Map



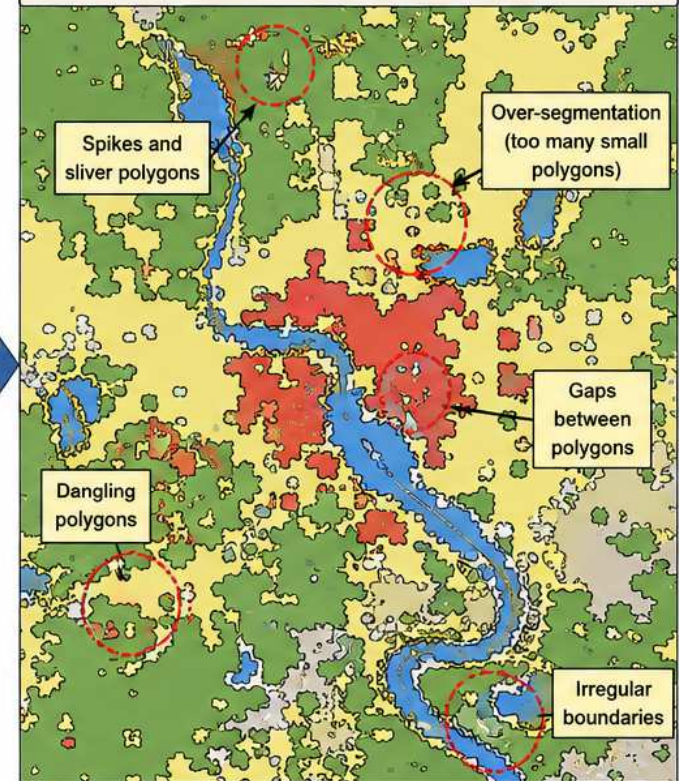
Forest Agriculture Built-up Water Barren

2. Vectorized Output
(Without Post Processing)



Direct vectorization of raster
(no cleaning or editing)

3. Problems in Vector Data
(Without Post Processing)



Key Point

Vectorization without post processing results in topological errors, noisy boundaries, slivers, gaps and over-segmentation. Post processing (cleaning, smoothing, editing) is essential for quality vector data.

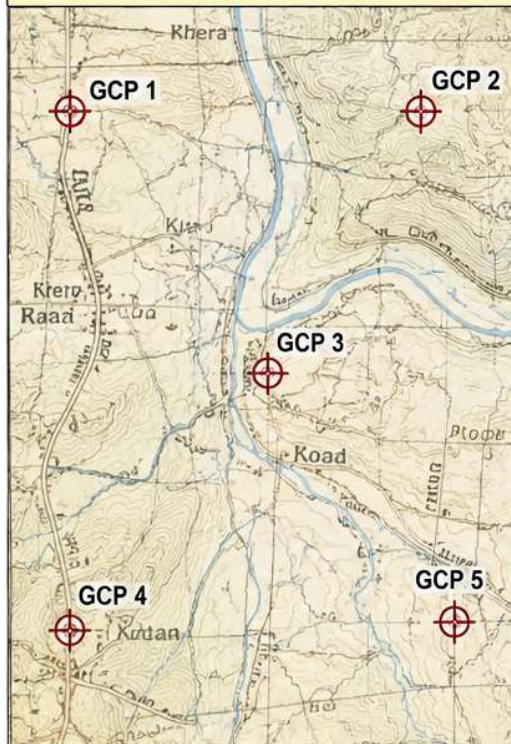
Common Issues

- Over-segmentation
- Spikes and sliver polygons
- Gaps and overlaps
- Dangling polygons
- Irregular boundaries
- Topology errors

3. Georeferencing with imprecise control points

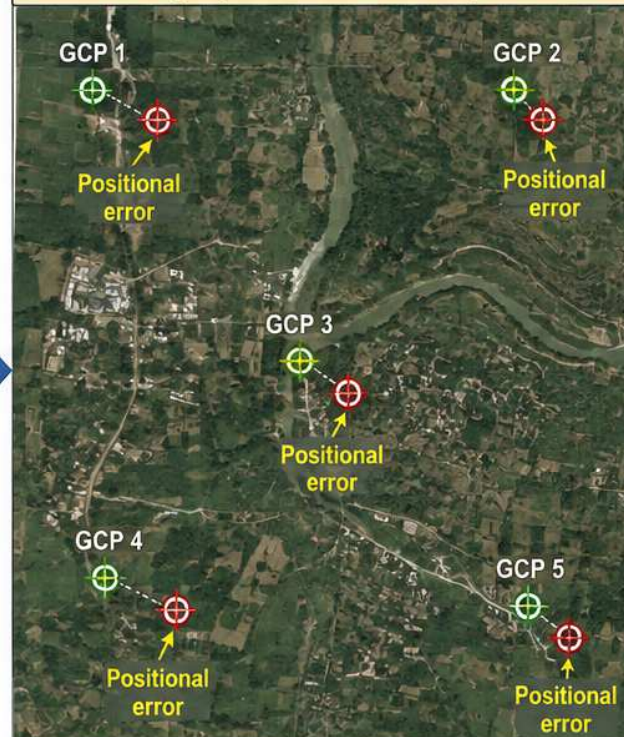
Georeferencing with Imprecise Control Points

1. Scanned Map (Source)



⊕ Selected Ground Control Points (GCPs)

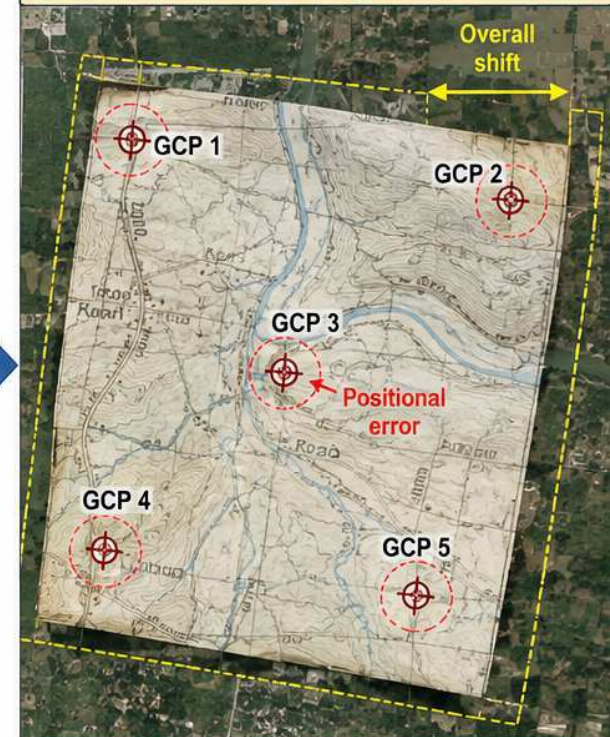
2. Control Points on Reference Map (Imprecise Selection)



⊕ True location (on reference map)

⊕ Imprecise GCP selected

3. Georeferenced Map (Result)



The map is misaligned due to imprecise control points, causing positional errors and distortion.

Key Point

Imprecise selection of Ground Control Points (GCPs) leads to geometric distortion, overall shift, and positional inaccuracy in the georeferenced map.

Legend

⊕ True location
⊕ Imprecise GCP

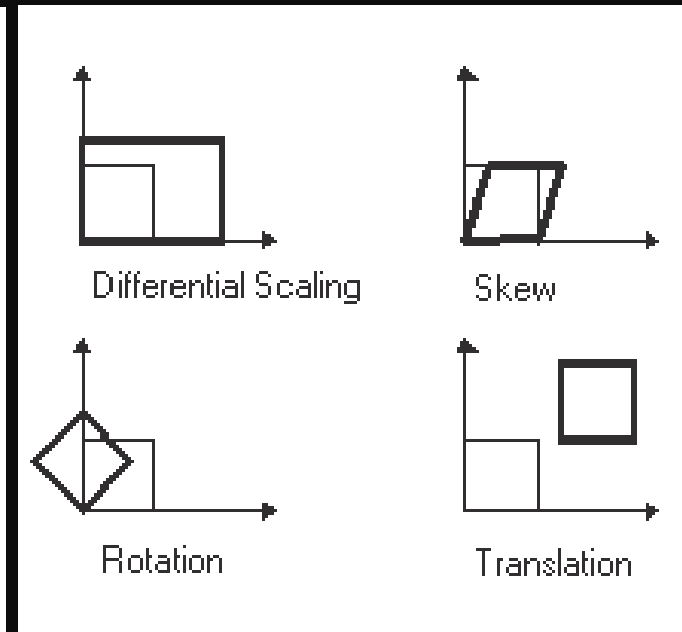
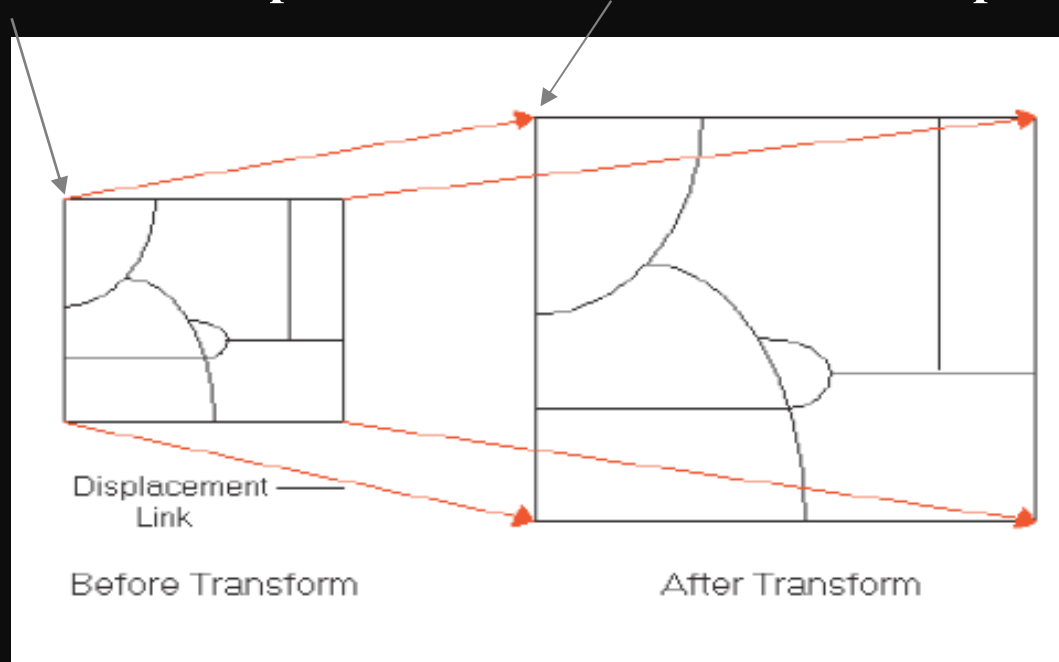
↔ Overall shift
--- Reference map extent

Transformation

Transformation converts data from one coordinate system to another; can be used to shift your data within a coordinate system (shift, rotate, enlargement and so on)

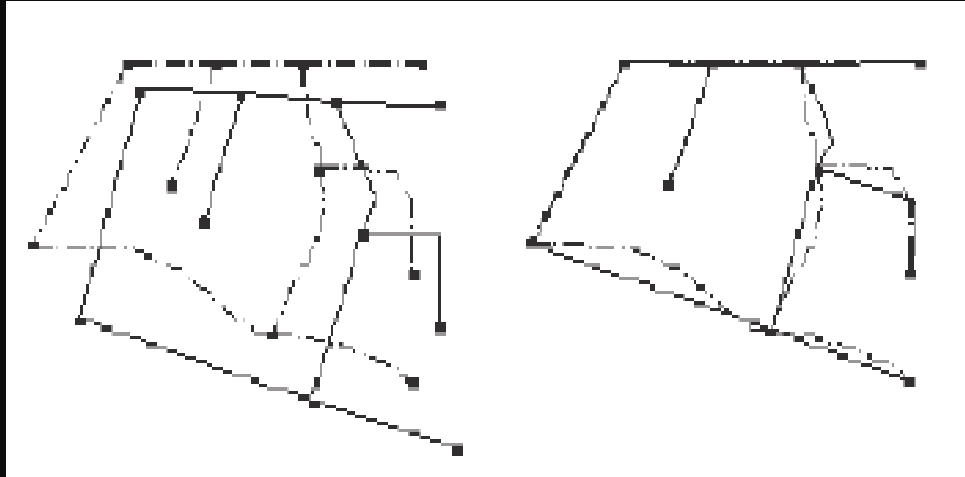
Source control point

Destination control point



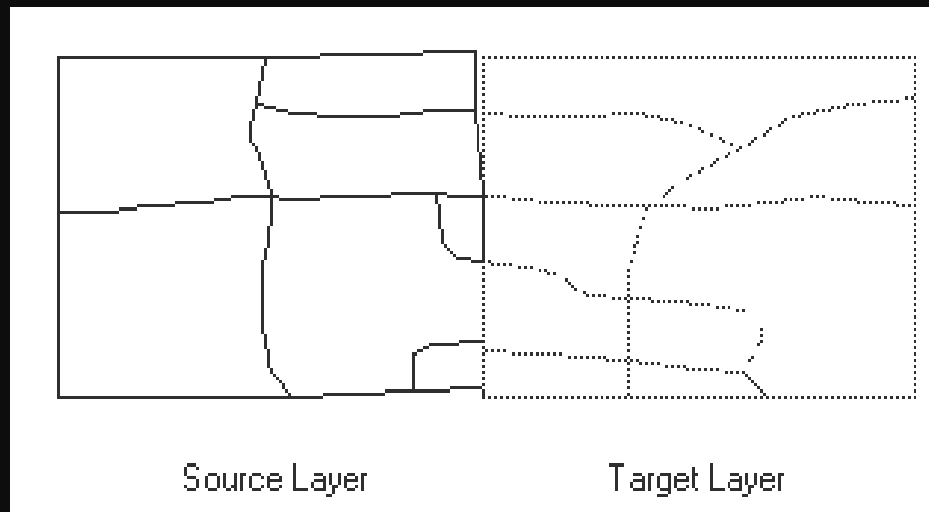
→ **Displacement links**

Rubber-sheeting



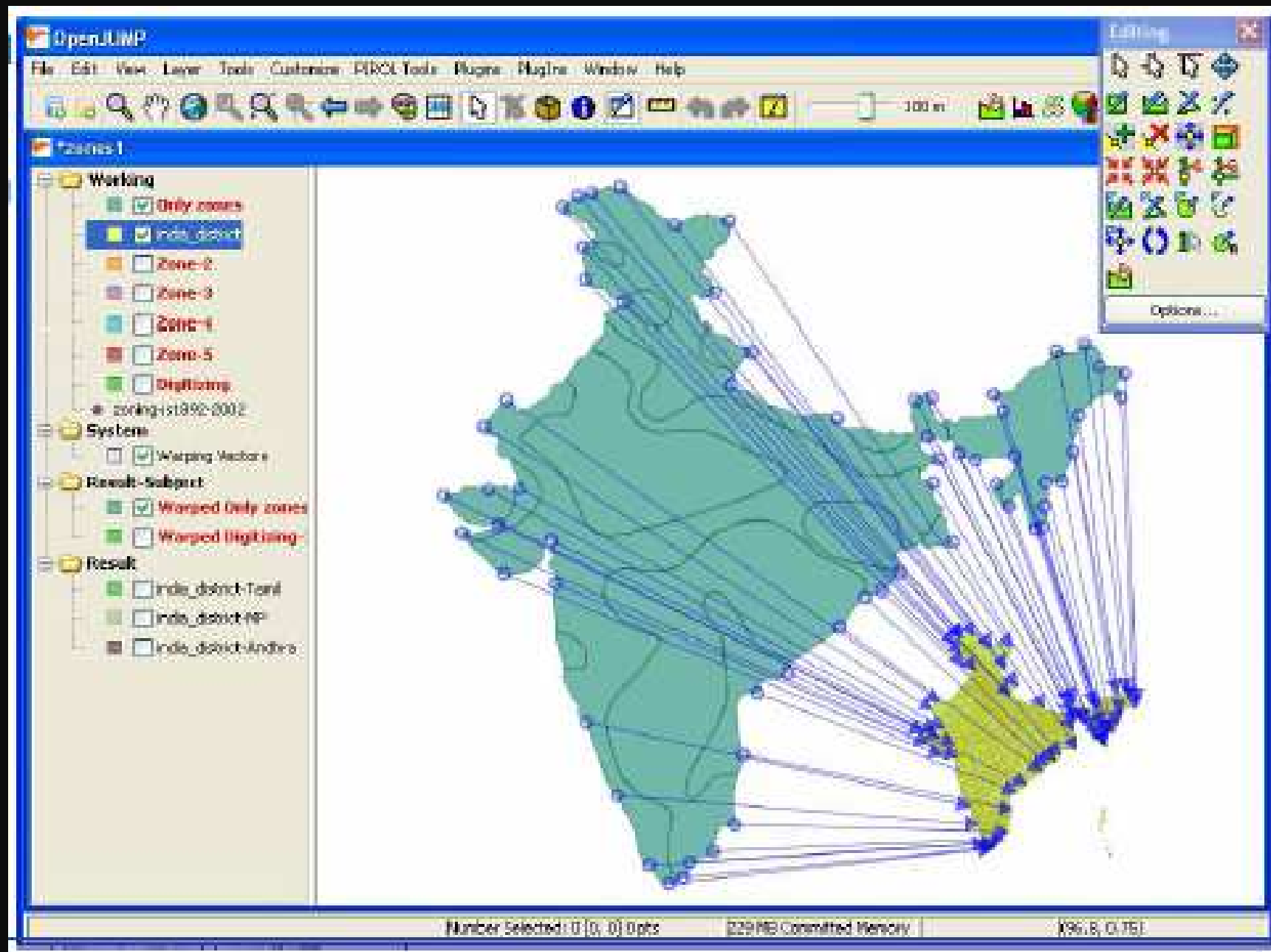
Rubber-sheeting is usually used to correct for geometric distortions; they may be introduced by imperfect registration in map compilation, lack of geodetic control in source data, and so on.

Edge-matching



- Rubber-sheeting adjusts source layer (dashed line) to target layer (solid line) based on displacement links
- Source layer is adjusted to target layer by setting snapping properties

Warping



4. Data entry errors

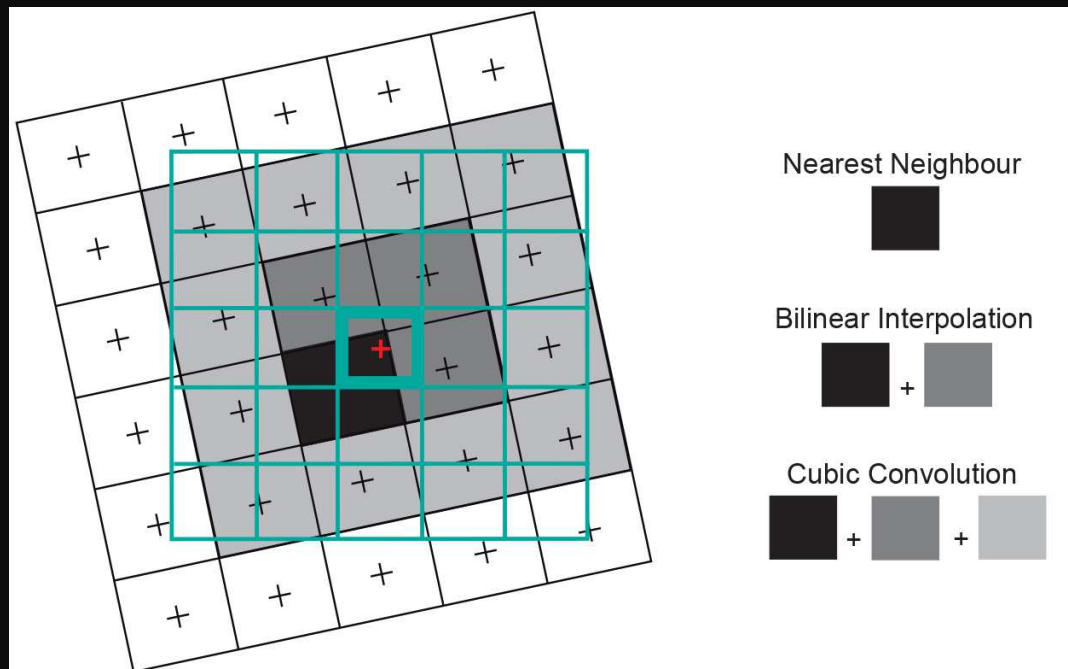
Missing attribute values

Compare the unique list of values to master list

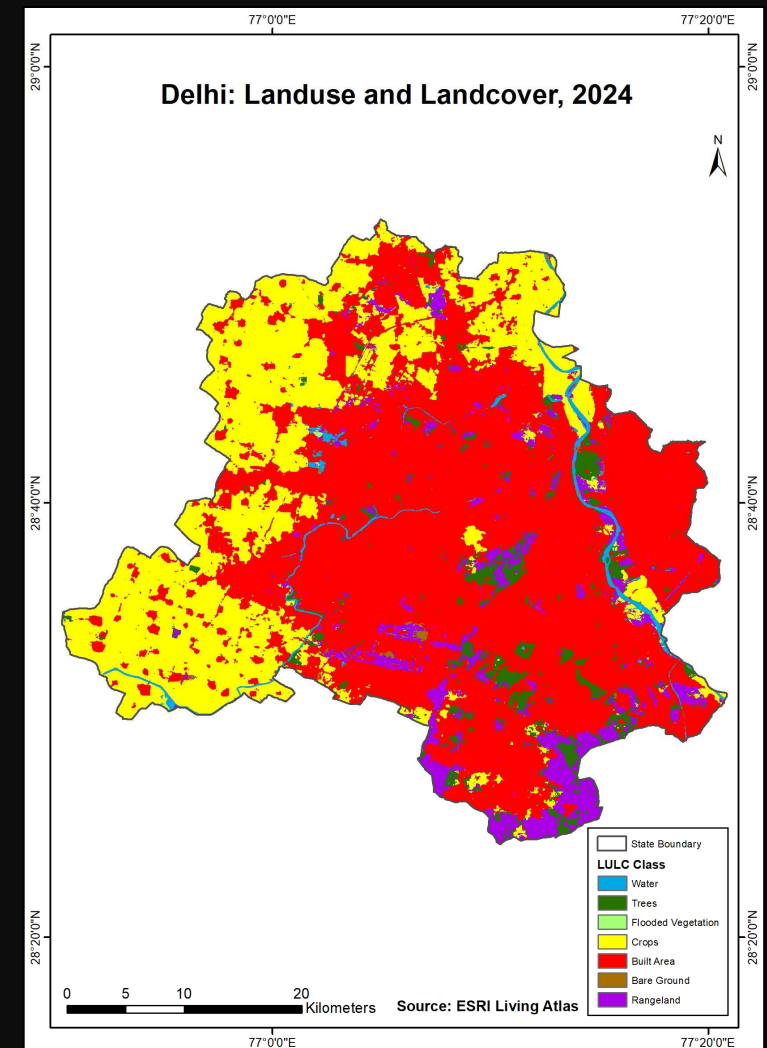
Incorrect attribute values

Typing errors: spelling check

5. Image interpretation error



ESRI Living Atlas LULC of Delhi



Data transfer:

information loss caused by
File conversion
Resampling of data

Object Relationships

- ◆ **Relationships between GIS objects**
 - ◆ Spatial relationships
 - ◆ Relationships as attributes

Feature relationships

- ◆ There are vast number of possible relationships in spatial data.
- ◆ Relationships are important in GIS analysis.
 - ◆ *"is contained in"*
 - ◆ *"intersects"*
- ◆ Relationships can exist between entities of the same type or of different types.

Types of relationship

Relationships which are used to construct complex objects from simple primitives.

- ◆ *Relationship between a line and the ordered set of points which defines it.*
- ◆ *Relationship between a polygon and the ordered set of lines which defines it*

Relationships which can be computed from the coordinates of the objects

- ◆ *Areas can be examined to see which one encloses a given point - the "is contained in" relationship can be computed.*
- ◆ *Areas can be examined to see if they overlap - the "overlaps" relationship*

◆ Relationships which cannot be computed from coordinates

We can compute if two lines cross, but not if the highways they represent intersect (may be an overpass).

Spatial relationships

